

### 6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let  $f(x) = x^2 - 4x - 12$ .

(a) Write  $f(x)$  in factored form, such that  $f(x) = (x - a)(x - b)$ . [2]

(b) Write down the  $x$ -intercepts of the graph of  $f$ . [1]

(c) Write down the  $y$ -intercept. [1]

(d) Write  $f(x)$  in vertex form, i.e.  $f(x) = (x - h)^2 + k$ . [1]

(e) Find the coordinates of the vertex of the parabola. [2]

|                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------|

2. [Maximum mark: 7]

Consider the function  $g(x) = -(x - 3)^2 + 16$ .

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that  $g(x) = -x^2 + 6x + 7$ . [2]

(d) Solve  $g(x) = 0$  to find the  $x$ -intercepts. [3]

3. [Maximum mark: 7]

The line  $y = 3x$  and the parabola  $y = x^2 - 4$  are drawn on the same axes.

(a) Show that the  $x$ -coordinates of the points of intersection satisfy

$$x^2 - 3x - 4 = 0. \quad [2]$$

(b) Show that  $x^2 - 3x - 4 = 0$  has two distinct real roots. [1]

(c) Factorise  $x^2 - 3x - 4$ . [1]

(d) Find the  $x$ -coordinates of the points of intersection. [1]

(e) Find the coordinates of both points of intersection as  $(x, y)$  pairs. [2]

|                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------|

4. [Maximum mark: 7]

The equation  $3x^2 - kx + 12 = 0$  has two equal roots.

- Write down the discriminant in terms of  $k$ . [1]
- Find the values of  $k$ . [2]
- For the positive value of  $k$ , find the repeated root. [2]
- State the values of  $k$  for which the equation has no real roots. [2]

### 6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let  $f(x) = x^2 + 6x - 16$ .

(a) Write  $f(x)$  in factored form, such that  $f(x) = (x - a)(x - b)$ . [2]

(b) Write down the  $x$ -intercepts of the graph of  $f$ . [1]

(c) Write down the  $y$ -intercept. [1]

(d) Write  $f(x)$  in vertex form, i.e.  $f(x) = (x - h)^2 + k$ . [1]

(e) Find the coordinates of the vertex of the parabola. [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

2. [Maximum mark: 7]

Consider the function  $g(x) = -(x - 5)^2 + 4$ .

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that  $g(x) = -x^2 + 10x - 21$ . [2]

(d) Solve  $g(x) = 0$  to find the  $x$ -intercepts. [3]

3. [Maximum mark: 7]

The line  $y = 4x$  and the parabola  $y = x^2 - 5$  are drawn on the same axes.

(a) Show that the  $x$ -coordinates of the points of intersection satisfy

$$x^2 - 4x - 5 = 0. \quad [2]$$

(b) Show that  $x^2 - 4x - 5 = 0$  has two distinct real roots. [1]

(c) Factorise  $x^2 - 4x - 5$ . [1]

(d) Find the  $x$ -coordinates of the points of intersection. [1]

(e) Find the coordinates of both points of intersection as  $(x, y)$  pairs. [2]

|                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> <p>.....</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------|

4. [Maximum mark: 7]

The equation  $2x^2 - kx + 32 = 0$  has two equal roots.

- Write down the discriminant in terms of  $k$ . [1]
- Find the values of  $k$ . [2]
- For the positive value of  $k$ , find the repeated root. [2]
- State the values of  $k$  for which the equation has no real roots. [2]